Bozhao Li **Robotics Software Engineer**

Target: NVIDIA Isaac Applications

Email libozhaovo@buaa.edu.cn
Phone +86 186 2233 2179
GitHub github.com/wee733

Core Skills

Platforms / Simulation: Isaac Sim / Isaac Lab / Isaac ROS, Jetson Thor / Orin, PyTorch / MuJoCo
Control / Engineering: RL / Sim2Real / Whole-Body Control, ROS 2, C++ / Python

Education

2024.09–Present	**Beihang University (BUAA)**	Cyber Science and Technology	M.S.
2020.09–2024.06	**Southwest University**	Computer and Information Science	B.Eng.

Experience

2025.11–Present **National Key R&D Program · B2 Quadruped-Manipulator Inspection Robot** — **Perception & Control Lead**

Led sim-to-real delivery of an **18-DoF whole-body RL controller**: reward and actuator modeling, the training-to-deployment contract, and a C++ real-time stack feeding Nav2 and manipulation.

- **Task reformulation:** no end-effector force sensor on hardware made force–position control unclosable; recast it as **coordinated base-velocity + EE-position tracking** in a 12-leg / 6-arm Isaac Lab PPO environment, preserving the observation contract to reuse pretrained weights.
- **Reward-hacking diagnosis:** the policy crouched on its hind legs to farm tracking reward on near-ground targets; offline reachability analysis **proved that pose cannot physically reach the target**, isolating a reward-design flaw rather than under-training — replaced it with **goal-conditioned pitch / roll / height supervision** and a one-sided height floor that closes the shortcut.
- **Sim2Real fidelity:** traced hardware arm oscillation to idealized training actuators; identified per-joint PD gains and latency, rebuilt them as **explicit PD models** with rate-equivalent coefficients across sim and hardware, and randomized friction / CoM / payload — **oscillation eliminated, stable walking under arm motion**.
- **Deployment contract:** one policy package and one FSM across **Isaac Lab, MuJoCo, and hardware** to stop interface drift; a **50 Hz inference / 1 kHz DDS-CAN** C++ runtime with dual-rate logs isolated joint-order, gait-phase, and zero-fallback bugs.
- **Drift correction:** the frozen policy observes no absolute heading and drifted out of Nav2 and manipulation tolerance; added a FAST-LIVO2 PI loop in command space **upstream of the untouched policy** instead of retraining.

2026.06–Present **Tsinghua Qiyuan Lab · Robot-Dog Swarm Control & Digital Twin** — **Isaac ROS Infra Lead**

Own low-level control, GPU perception, and ROS 2 infrastructure for B2 + R5A (**Isaac Sim 5.1 / Isaac ROS 4.5 / AGX Thor**), **exposing a stable interface** to the VLN team's segmentation, VLA, and decision models.

- **Hardware into Isaac ROS:** the unsupported ARX R5A blocked the stack; wrote a **ros2_control plugin** and URDF / SRDF / XRDF collapsing vendor CAN into a standard interface — MoveIt 2 and cuMotion now drive hardware **without modifying upstream source**.
- **Unified sim/real interface:** official **ZED X / XT32 / D435i** assets in a **single core USD** emit driver-identical topic / frame contracts via the **OmniGraph ROS 2 Bridge**, bypassing the Isaac Lab / ROS 2 runtime conflict; splitting obstacles into **depth vs. lidar modes** over one planner **narrows navigation faults to perception or planning**.
- **Task orchestration:** standardized perception into target-pose / obstacle interfaces feeding a custom **cuMotion pipeline**; reworked **Isaac ROS Manipulation behavior trees** so low-rate semantic commands drive local planning and closed-loop door-opening and debris clearing.
- **Synthetic data & arm–leg decoupling:** fixed the base in Isaac Sim to mass-generate collision-free **cuMotion** trajectories as a synthetic manipulation dataset; fine-tuned **GR00T N1.7** and explored **arm–leg decoupling** in which RL treats arm motion as an **observable disturbance**, so the base compensates the induced CoM and torque shifts.
- **Fast model turnaround:** first **Thor deployment** of LocateAnything-3B replacing YOLO, plus a fine-tuned **DINOv3** — a same-day model-to-robot loop.

2026.05–2026.07 **DongYi Tech · Phybot C2 Humanoid Soccer & Sim2Real** — **RL & Sim2Real Engineer**

Delivered one **Isaac Lab–MuJoCo–hardware** path for four skills on a **21-DoF humanoid**: walking, penalty kick, soccer kick, drop kick.

- **Multi-skill policies & deployment:** trained walking and kicking with **AMP / PPO** behind one policy entry, adding reference-motion residual control and dedicated FSMs for dynamic kicks; diffed training against the C++ deployment to trace hardware misbehavior to **conflated yaw-free vs. yaw-aware pose semantics**, and validated all four skills at **500 Hz motor / 50 Hz policy**.

2025.03–2026.04 **China Aerospace 9th Academy · 30-UAV Swarm** — **Swarm & Onboard Software**

Owned Jetson computing, mission planning, and communications for a **30-UAV swarm**; led the ROS 1-to-ROS 2 migration and field validation.

- **Onboard system & swarm planning:** integrated PX4, MAVROS, and visual / inertial sensors on **Orin NX** and led the **ROS 1 to ROS 2** migration; implemented 3D space-time weighted A* with conflict resolution and replaced DDS with **Zenoh** against cross-subnet discovery failures, validating **public-5G swarm control** in the field.
- **Edge app management:** built an onboard **robot application manager** (FastAPI / Vue 3) closing a **cloud-publish to one-click-deploy** loop with unified **Foxglove** telemetry, deployed fleet-wide.

2024.11–2025.10 **Beijing Zhuoyi Tech · Hangfeng Autonomous UAV** — **Deputy Chief Designer**

Owned electrical design, assembly, and mechanical iteration from system design through flight test; integrated **FAST-LIO2 / EGO-Planner / RACER / D2SLAM**, unified timing, frames, and state-machine interfaces, and built log-based regression for drift, oscillation, and planning timeouts.

Publications

Title	Venue	Role	Status
Odometry-Guided Command-Space Drift Correction for a Frozen RL Quadruped-Arm Policy	ICARCV 2026	First author	Submitted
Security-Enhanced Formation Tracking Control for Networked Swarm Systems via Quantized Encryption-Decryption	IEEE TCNS	Co-first author	**Accepted**
FormationDiffuser: Boundary-Consistent Diffusion for Long-Horizon 3D Multi-UAV Reconfiguration	AAAI 2027	Co-first author	Submitted
IRA: Interception-Risk-Aware Trajectory Planning for MAV Evasion from Multiple Pursuers in Cluttered Environments	IEEE RA-L	Second author	Under Review